

Chapter 107 — Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Pdet = Pves – Pabd**; Pabd by rectal catheter preferred.
- **Storage/DLPP > 40 cm H₂O** → upper-tract damage (McGuire); aim as low as achievable.
- **ALPP < 60 = ISD (intrinsic sphincter deficiency)**; 60–90 equivocal; ≥ 90 little/no ISD.
- **BOOI (bladder outlet obstruction index) = PdetQmax – 2Qmax**: > 40 obstructed, 20–40 equivocal, < 20 unobstructed.
- **BCI (bladder contractility index) = PdetQmax + 5Qmax**: > 150 strong, 100–150 normal, < 100 weak.
- **VUDS (videourodynamics) = gold standard** in NLUTD (neurogenic lower urinary tract dysfunction).
- **PBNO (primary bladder neck obstruction) and DISD** diagnosable **only on VUDS**.
- **True DESD needs suprasacral lesion**; no lesion → dysfunctional voiding.
- **VALUE**: office evaluation alone **non-inferior** before SUI surgery (76.9% vs 77.2%).
- **Impaired compliance = only absolute urodynamic indication** to treat BOO.

ROLE OF URODYNAMICS

- **Use UDS when**: diagnosis unclear after history/exam/simple tests, or disease threatens upper tracts silently.
- **Box 107.1 risk factors**: impaired compliance · DESD (detrusor-external sphincter dyssynergia) · DISD (detrusor-internal) · high-pressure DO (detrusor overactivity) throughout filling · **DLPP > 40** · poor emptying + high storage pressure.
- **Treat Box 107.1 findings** even if asymptomatic.

CONDUCTING THE STUDY

- **Wein functional classification (1981)**: storage vs emptying × bladder vs outlet.
- **Overactive bladder / underactive outlet** → storage failure; **underactive bladder / overactive outlet** → emptying failure.
- **Three rules**: questions first → design → customise.
- **Drug effect question** → test on drug; **aetiology question** → consider stopping.
- **Pre-invasive**: symptom score, **3-day frequency-volume chart**, free flow, PVR (postvoid residual).
- **Urinalysis before every study**; **symptomatic UTI** → postpone.
- **Prophylaxis**: single dose ≤ 1 h before; **first-line oral trimethoprim-sulfamethoxazole**.
- **Prophylaxis yes**: NLUTD · BOO/raised PVR · **age > 70** · asymptomatic bacteriuria · immunosuppression/renal transplant · catheters/CIC · at-risk joint implant.
- **Prophylaxis no**: normal anatomy (**level 1**) · diabetes · recurrent UTI · menopause · gender · obesity · prior surgery · valve disease.

COMPONENTS

- **Voiding efficiency = VV / (VV + PVR) × 100**.
- **Raised PVR**: confirms poor emptying, not cause → repeat.
- **Flow curves**: bell = normal; **flattened = fixed obstruction**; interrupted = underactivity/straining.
- **Adult fill rate 30–100 mL/min**.
- **EMG standard**: two perineal **surface** electrodes.
- **Urethral pressure** = pressure just to open closed urethra.

EQUIPMENT

- **Water-filled**: zero at **symphysis pubis**, no air, motion artefact; basis of most nomograms.
- **Air-charged**: no levelling/flushing; **not interchangeable** (Pdet differs up to **10 cm H₂O**).
- **Microtip**: expensive, reusable, sterilise.
- **Infusate**: saline/contrast; **never CO₂**.
- **Flowmeters**: **load cell** (needs urine density) and **rotating disc** commonest.

- **Needle EMG** best signal; **surface EMG** most comfortable.

ANALYSIS AND INTERPRETATION

- **Normal filling**: Pdet ≈ 0 throughout, no IDCs (involuntary detrusor contractions).
- **DO subtypes identical on trace** — named by neurological history (IDO, NDO, non-neurogenic).
- **No DO on UDS in up to 50%** of women with UUI.
- **Single terminal IDC + BOO** → better OAB resolution after TURP.
- **DO commoner upright** → test in symptomatic position.
- **Compliance = $\Delta V / \Delta P_{det}$** ; < 20 mL/cm H₂O proposed impaired; absolute pressure more useful.
- **Pop-offs**: VUR, diverticulum, incompetent outlet.
- **ALPP**: total pressure, from **150 mL**, then every **50 mL**; remove catheter if no leak.
- **VLPP < CLPP**; **larger catheter** → **lower ALPP**.
- **DLPP 20 cm H₂O** more sensitive than 40 in myelodysplasia.
- **Hypermobility**: Q-tip > 30°.
- **MUCP ≤ 20 cm H₂O = ISD**; continent women MUCP **40–60**, functional length ~3 cm.
- **Minimum VV 150 mL**; corrected Qmax = $Q_{max} / \sqrt{VV}$.
- **Bethanechol test**: **2.5 mg SC**; positive ≥ **15 cm H₂O** rise at 30 min — unreliable.
- **Micturition sequence**: sphincter relaxes → detrusor contracts → neck opens → flow.
- **PFS patterns**: high P/low Q = obstruction; low P/low Q = underactivity.
- **Stop test**: voluntary gives **lowest Piso**.
- **Piso < 50 cm H₂O (men)** = detrusor underactivity.
- **Obstruction hidden** if Pdet cannot reach 40.
- **Watts factor**: $a \approx 25 \text{ cm H}_2\text{O}$, $b \approx 6 \text{ mm/s}$; $W_{max} 7.0 \text{ W/m}^2$.
- **MUPP (Yalla)**: prostatic drop > **5 cm H₂O** = obstruction.
- **Female BOO cut-points**: $Q_{max} \leq 15 / \leq 11 / \leq 12$, $P_{det}Q_{max} \geq 20 / \geq 21 / \geq 25$.
- **Solomon-Greenwell**: $P_{det}Q_{max} - 2.2Q_{max} < 0 \rightarrow < 10\%$; $> 5 \rightarrow > 50\%$; $> 18 \rightarrow > 90\%$.
- **Blaivas-Groutz** overdiagnoses; uses free flow + Pdetmax.
- **DISD**: lesion above **T10-L1**.

VIDEOURODYNAMICS

- **Positions**: SUI standing; voiding — women sitting, men standing.
- **Screen only**: provocation, pressure rises, voiding.
- **PBNO types**: I high P/low Q · II normal P/low Q · III delayed opening.
- **VUR**: record volume + pressure at onset; masks poor compliance.

- **EAU:** VUDS optimum; unavailable → **CMG into PFS**, physiological fill, warm saline.
- **Refer to VUDS centre:** NLUTD, unexplained female retention, radical pelvic surgery, diversion, transplant, pelvic radiation.

CLINICAL APPLICATIONS

- **Oxford grades:** A level 1 · B level 2/3 · C level 4/expert opinion.
- **SISTER:** UDS (ALPP, DO) did not predict Burch/sling outcome.
- **AUA SUI 2023:** may omit UDS in index patient (conditional, grade B).
- **Test with catheter removed** if SUI not shown.
- **High-grade POP, no SUI** → stress test **with reduction**.
- **NICE 2006 pre-SUI UDS if:** suspected DO · prior SUI/anterior prolapse surgery · voiding dysfunction.
- **EAU:** no routine UDS in uncomplicated SUI; **no UPP/LPP for severity**.
- **AUA/SUFU:** PFS in men before invasive treatment — **Standard, grade B**.
- **EAU PFS triggers:** $VV \leq 150 \text{ mL}$ · $Q_{\max} > 10$ · $PVR > 300$ · age > 80 or < 50 · failed treatment.
- **No UDS** before α -blocker or for uncomplicated OAB.
- **Storage symptoms persist** $\sim \frac{1}{2}$ after TURP.
- **NLUTD UDS essential:** SCI, myelomeningocele; **not initially:** post-stroke UUI, MS with low PVR.
- **Ice water test (Bors-Blinn 1957):** UMN positive, LMN negative; never before standard UDS.

FUTURE DIRECTIONS

- **Filling cystometry** remains **gold standard** for storage disorders.
- **Penile cuff test:** catheter-free, naturally filled bladder, modified ICS nomogram.

CLASSIC TRAPS

- **Trap:** low ALPP and high DLPP mean the same — **Actually:** weak sphincter vs dangerous outlet.
- **Trap:** EMG rise with IDC = DESD — **Actually:** normal guarding reflex.
- **Trap:** sphincter dyssynergia without neurological lesion = DESD — **Actually:** dysfunctional voiding.
- **Trap:** BOOI/ICS nomogram in women — **Actually:** grossly underestimates female BOO.
- **Trap:** low compliance number always dangerous — **Actually:** small bladders mislead; judge absolute pressure.
- **Trap:** low end-fill pressure = safe bladder — **Actually:** VUR pop-off may hide poor compliance.
- **Trap:** EAU PFS if $Q_{\max} < 10$ — **Actually:** trigger is $Q_{\max} > 10$.
- **Trap:** recurrent UTI or valve disease needs prophylaxis — **Actually:** no; asymptomatic bacteriuria yes.
- **Trap:** failed void in lab = acontractile detrusor — **Actually:** "situational inability to void as usual".
- **Trap:** VLPP and CLPP interchangeable — **Actually:** VLPP significantly lower.